Where the Earth's plates grind

45 years of M≥5 earthquakes from the USGS ANSS catalog, 1980–2025. Drag the year and magnitude controls, switch the projection focus, and click a hex on the map to inspect the largest events in that region.

Controls

Start year

1980

End year

2025

Minimum magnitude

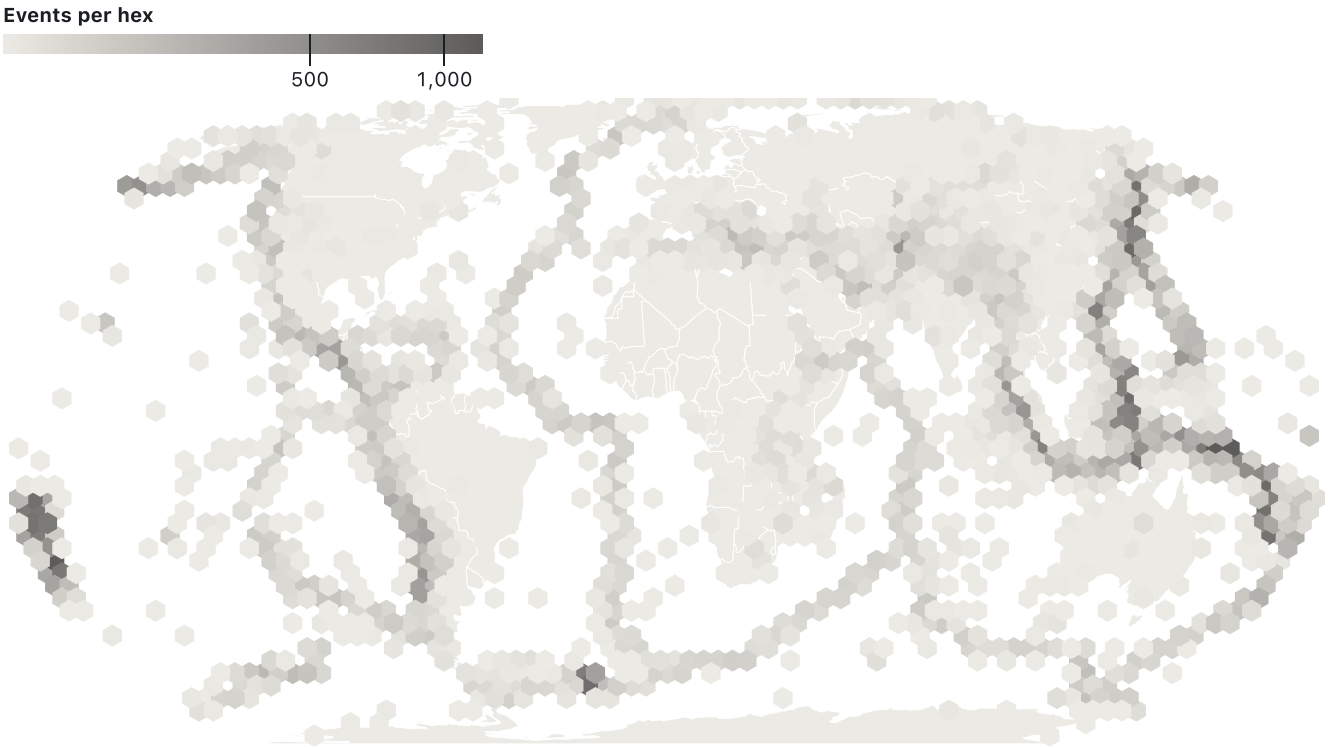
5

Region

World

Reset all filters

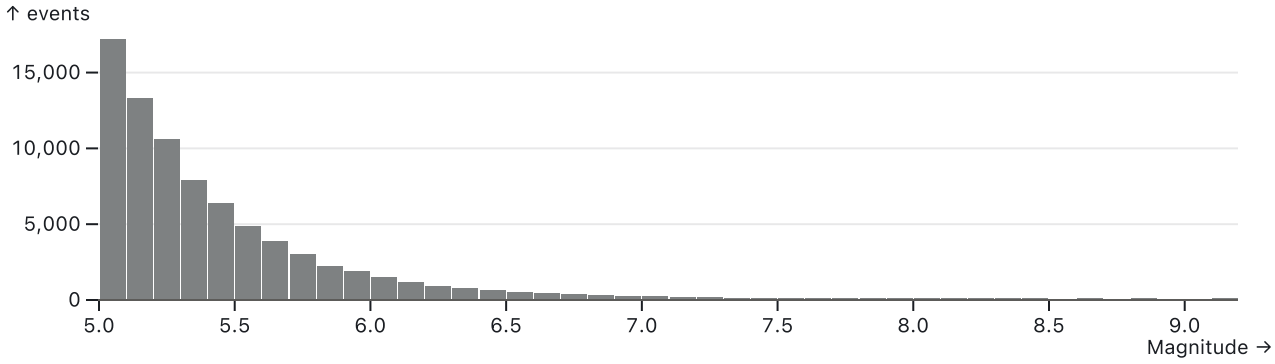
Map



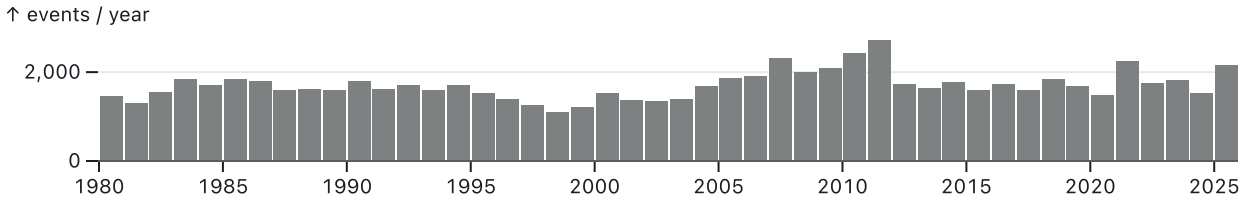
Selected hex — top events

Click a hex on the map to inspect the largest events in that region.

Magnitude distribution



Events per year



Data. USGS Earthquake Hazards Program, Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat). Queried via the FDSN event web service: earthquake.usgs.gov/fdsnws/event/1/. All events with magnitude ≥ 5.0 between 1980-01-01 and 2025-12-31 (~75,000 events). USGS preferred magnitudes are heterogeneous across the catalog (Mw, mb, ML, Ms); for visualization purposes the catalog is treated as a single magnitude axis. Pre-1980 catalog completeness in remote ocean basins is materially lower than post-1990 due to broadband seismograph network expansion — starting at 1980 avoids that caveat without losing analytical content. A small fraction (~2%) of place names from the upstream feed contain a literal "?" where a non-ASCII character was lost during USGS ingestion (e.g. "?funato" for "Ōfunato"); these are preserved unchanged here.

Basemap. Natural Earth countries, 110m resolution, decoded from world-atlas TopoJSON via topojson-client. Equal-Earth projection.

Tooling. Built with Observable Framework, Observable Plot, and D3. Cell scaffold and iteration assistance from Claude Code (Opus 4.7, with the Playwright and Context7 MCP servers). Deployed to GitHub Pages via Actions.